

# The Cooperation and Competition Variant Boundary

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**Date:** May 15, 2026

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## Attribution

*This work derives from and formalizes the inter-Self coordination architecture introduced in “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026), the third paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026) and “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026).*

*It does not introduce new axioms. Its sole contribution is to draw the boundary between the cooperation and competition variants of a Full Aspect Integration (FAI) event with definitional precision, so that downstream work can adopt or argue against the distinction without ambiguity.*

## Abstract

The CKS inter-Self coordination architecture provides two variants of a FAI event — cooperation and competition — that share the same shared-substrate primitive but operate under different conflict-routing rules. The cooperation/competition variant is a configurable dimension of the FAI event configuration: it is substrate content, authored under governance, and carries the same inspectability and modifiability as any other governance object. This note formalizes the boundary between the two variants by answering the question that makes the boundary sharp: what determines which variant an event instantiates, and can an event’s effective variant change without an explicit governance configuration amendment? The answer is that the variant is determined by the authored Dimension 4 configuration, not by the event’s conflict density, joint-output volume, or any other outcome characteristic. An event cannot drift from cooperation to competition — or in the reverse direction — without governance amending both the Dimension 4 entry and the conflict-routing rules. When a cooperation-configured event produces unexpectedly competition-like outcomes, that mismatch is a post-mortem finding that informs future configuration decisions; it is not evidence that a third, “mixed” variant exists.

## 1. Why the boundary requires precise statement

The cooperation and competition variants of a FAI event are distinguished in the architecture by their conflict-routing posture: cooperation-variant events use a resolve-tier-dominant conflict-routing configuration, optimizing shared-substrate orchestration for mutual benefit;

competition-variant events use a preserve-tier-dominant configuration, scoping contribution and protecting strategically sensitive content within the shared substrate. The distinction matters architecturally because the two routing postures produce structurally different shared substrates. It also matters for governance accountability: when a FAI event's conflict-handling record is later inspected, the routing posture the configuration specified determines what the event was supposed to do, and therefore what a dense or sparse conflict registry means.

Without a precise answer to what determines the variant, two problems arise. First, an event with mixed outcomes — joint outputs produced alongside a dense conflict registry — could be claimed to instantiate a third, “mixed” variant not specified by the architecture. This claim would undermine the governance-substrate-content principle that all configurable dimensions of a FAI event are authored under governance: a “mixed” variant that emerges from outcome rather than configuration is a variant outside governance control. Second, governance actors could rationalize post-hoc variant reassignment — treating an event as a competition event because it “functioned like” one — without the configuration amendment the architecture requires. This rationalization produces governance theater: the substrate's configuration record says one thing; the routing behavior or post-hoc classification says another.

Both problems are closed by the same principle. This note formalizes that principle.

## 2. The boundary scenario

Three FAI events with different characteristics present the boundary:

**(a) Clear cooperation.** Governance explicitly configured the cooperation variant in the FAI event configuration (Dimension 4). Conflict routing is resolve-tier-dominant. The event produces joint outputs; no comparative benchmarking was intended. The conflict registry is sparse relative to the shared-substrate content exchanged.

**(b) Clear competition.** Governance explicitly configured the competition variant. Conflict routing is preserve-tier-dominant. The event produces comparative intelligence; no joint outputs were intended. The conflict registry is dense, preserving the strategic divergences between participating Selves.

**(c) Ambiguous.** Governance configured the cooperation variant. Conflict routing is resolve-tier-dominant as specified. But the event produces an unexpectedly dense conflict registry. Joint outputs are minimal. A post-mortem review finds that the event functioned more like a competition event — the shared substrate accumulated preserved conflicts rather than resolved ones, and the outputs resemble comparative intelligence rather than cooperative production.

The question the boundary case poses: is scenario (c) a cooperation event or a competition event?

### 3. The architecture's resolution: the variant is authored configuration, not outcome

The cooperation/competition variant is Dimension 4 of the FAI event configuration. The FAI event configuration is substrate content — it is authored under the joint governance authority of participating Selves, it is inspectable at any time, and it is modifiable only through governance action. This is the configuration-as-substrate-content commitment that Paper 3 (§8) develops: every configurable dimension of the FAI mechanism, including the cooperation/competition variant, lives as substrate content within the governance architecture, with the same authority structure as other substrate content.

What an event *produces* — the volume of joint outputs, the density of the conflict registry, the character of the content that propagates to participating Selves at dissolution — is outcome. Outcome reflects how well the configuration fit the actual coordination task. It does not have definitional authority over the variant. The governance-authoritative source for the variant is the authored configuration object, not the pattern of outputs.

Applying this principle to the three scenarios:

**(a) Clear cooperation.** Dimension 4 of the configuration specifies cooperation variant. The event is a cooperation event. The sparse conflict registry is consistent with the configuration's intent. No ambiguity.

**(b) Clear competition.** Dimension 4 specifies competition variant. The event is a competition event. The dense conflict registry is the expected preserve-tier output. No ambiguity.

**(c) Ambiguous.** Dimension 4 specifies cooperation variant. The event **is a cooperation event**. The dense conflict registry and the minimal joint outputs do not change this. The event was cooperation-configured; it ran under resolve-tier-dominant routing as that configuration requires; what it produced reflects the fitness of the cooperation variant for the actual sharing scope and domain relationship — not the variant's identity. The post-mortem finding is that the cooperation variant was a poor choice for this event, not that the event was a competition event.

Scenario (c) is, precisely: a cooperation event with poor governance fit.

### 4. What variant drift without configuration amendment would mean

Suppose governance, observing scenario (c)'s dense conflict registry mid-event, decides that competition variant would be more appropriate and directs the shared substrate's orchestration to shift toward preserve-tier-dominant routing — without amending Dimension 4 of the configuration.

This produces governance theater. The configuration record of the event specifies cooperation variant; the actual routing behavior instantiates competition-variant posture. The two authoritative

sources — the authored configuration (cooperation) and the operational routing pattern (preserve-tier-dominant) — assert contradictory things about what the event is. The substrate's inspectability and the path-retraceability of its conflict-handling record are undermined: a downstream review of the event's configuration cannot reconstruct what routing logic was actually applied.

The only governance-compliant response to the mid-event finding that competition variant would be more appropriate is a configuration amendment: governance amends Dimension 4 to competition variant AND updates the conflict-routing rules to preserve-tier-dominant. The amendment is the governance mechanism for legitimate variant change. Without the amendment, the event remains cooperation-variant regardless of its effective routing behavior. With the amendment, the variant change is recorded as substrate content, the routing shift is traceable to the amendment, and the post-event configuration record accurately describes what the event was and when it changed.

The amendment requirement is not a procedural formality. It is the architectural property that keeps the variant definition anchored in governance rather than in outcome. Allowing outcome to redefine the variant — even well-intentioned outcome observation — converts the Dimension 4 configuration from a governance-authoritative object into a default that operational behavior can silently override. That conversion is incompatible with the configuration-as-substrate-content commitment.

## 5. The governance implication of outcome mismatch

When a cooperation-configured event produces competition-like outcomes, the mismatch is governance intelligence. It is the raw material for three post-mortem inquiries:

**Sharing scope review.** Was the sharing scope too broad for cooperation purposes? The aspects contributed may have been sufficiently differentiated that resolve-tier routing could not produce genuine joint outputs — the divergences were too fundamental for orchestration to bridge. A narrower sharing scope, or a different selection of contributed aspects, may have produced a cooperation-consistent outcome.

**Partner-relationship calibration.** Does the pattern of preserved conflicts suggest that this partner relationship is better served by competition-variant events? The conflict registry records what the Selves' contributed aspects disagreed about. If those disagreements are structural — if they recur across domains and aspects — the relationship may be one in which comparative intelligence is the more honest and useful output, and competition variant is the appropriate configuration for future events.

**Orchestration rule coverage.** Were the conflict classes that drove the dense registry covered by the event's orchestration rules? Dense preserve-tier output in a resolve-tier-dominant event sometimes reflects not a variant mismatch but an orchestration-rule gap: the rules did not cover the conflict classes that the contributed aspects generated. Expanding the orchestration

rules to cover those classes may produce a future cooperation event that functions as intended.

In all three cases, the dense conflict registry is useful governance information precisely because the event is unambiguously identified as a cooperation event with poor fit. If the event were reclassified as a competition event post-hoc, the three post-mortem inquiries would be suppressed: there would be no governance finding to act on, because the outcome would appear to have matched the (reassigned) variant. The misconfiguration would persist undetected into future events.

Keeping the variant anchored in the authored configuration preserves the feedback loop between outcome and future governance decisions. That feedback loop is the mechanism through which post-mortem review improves future FAI event configurations.

## **6. Prior-art significance**

The principle that the authored configuration determines the variant — not the event’s outcome — forecloses a specific adversarial claim: that FAI events with mixed characteristics (some cooperation-consistent, some competition-consistent) constitute a third architectural variant. This claim would be significant if it were available, because a “mixed” variant outside the cooperation/competition taxonomy would be a variant outside the governance-configured dimensions of the architecture — and therefore a coordination mode that governance does not author, inspect, or amend.

The configuration-as-variant-determiner principle closes this claim at the definitional level. Every FAI event has an authored Dimension 4 value; that value is cooperation or competition; there is no third substrate option. Mixed outcomes reflect configuration fitness, not configuration plurality. The boundary test below operationalizes this for any event.

A separate adversarial reading would have an event’s effective routing behavior — as opposed to its outcome characteristics — redefine its variant. This reading is closed by the amendment requirement: effective routing behavior is determined by the conflict-routing rules in the event’s configuration substrate; those rules are themselves substrate content; changing them requires a configuration amendment. An event with resolve-tier-dominant routing rules and a dense preserve-tier conflict registry has routing rules that did not produce the expected output — it does not have competition-variant routing rules.

## **7. Operational test**

For any FAI event, the cooperation/competition variant boundary is determined by the following test:

1. Locate the event’s Dimension 4 configuration entry in the FAI event configuration.
2. Read the value: cooperation or competition.

3. That value is the event's variant.

If the Dimension 4 entry specifies cooperation variant and the event's conflict registry is dense, the event is a cooperation event with a governance-quality finding. The finding is: the cooperation variant was a poor fit for this coordination task; governance should consider whether future events with this partner, in this domain, should use competition variant.

If the Dimension 4 entry specifies competition variant and the event produces joint outputs, the event is a competition event. The finding is: the competition variant may have been unnecessarily restrictive; governance should consider whether cooperation variant would have produced comparable comparative intelligence with greater mutual benefit.

If the Dimension 4 entry was amended mid-event, the event has two variant periods: the pre-amendment period under the original variant, and the post-amendment period under the amended variant. Both periods are traceable in the configuration substrate's amendment history.

There is no governance-compliant path by which an event's variant is determined by its outcomes rather than its configuration. A system that allows outcome-determined variant assignment does not instantiate the configuration-as-substrate-content commitment.

## **8. Conclusion**

The cooperation/competition variant boundary is sharp: the variant is what Dimension 4 of the FAI event configuration specifies, and it is determined at configuration time under governance authority. Outcome characteristics — conflict density, joint-output volume, the character of evolution-feed content — reflect how well the configuration fit the task; they do not redefine the variant. An event cannot drift from one variant to the other without a configuration amendment that is itself substrate content, traceable, and inspectable under the same governance authority that authored the original configuration.

When a cooperation-configured event produces competition-like outcomes, the governance-compliant response is to treat the mismatch as a post-mortem finding and use it to improve future configurations — not to reclassify the event. Reclassification suppresses the feedback loop that makes post-mortem review valuable. Configuration-anchored variant identity preserves that feedback loop and keeps the governance record of FAI events accurate.

Subsequent work that extends or composes this architecture should treat the cooperation/competition variant as an authored governance object with no outcome-determined alternative. Work that proposes outcome-based variant classification is proposing a different governance architecture, and the difference should be named.

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**Source paper**

Li, W. (2026). *Inter-Self Coordination via Shared Substrate / Full Aspect Integration*. April 2026. ORCID: 0009-0004-8065-3235.

**How to cite this note**

Li, W. (2026). *The Cooperation and Competition Variant Boundary*. May 15, 2026. ORCID: 0009-0004-8065-3235.